

Machine Learning Techniques to Improve Accuracy in Cardiovascular Disease Prediction

Introduction

Cardiovascular disease (CVD) is a leading cause of death globally, accounting for over 17.9 million deaths each year, nearly a third of all deaths worldwide. Many cases of CVD, including coronary artery disease and heart failure, are preventable, current challenges around early diagnosis and risk prediction complicate effective treatment and patient outcomes. As health data and computing power have grown, the application of machine learning (ML) provides an opportunity to enhance diagnostic accuracy and support clinicians in managing CVD.

This report reviews, *"A Comparative Analysis of Machine Learning Techniques and Key Insights for Cardiovascular Disease Prediction"*, which presents a framework using multiple machine learning classification techniques. Focusing on the technical approaches, methodological choices, and the practical relevance of findings, to develop a predictive model for early detection and management of heart disease. (Hossain, 2024)

Background and Literature Review

The broad category of cardiovascular diseases includes conditions from valvular and atherosclerotic disease to arrhythmias and cardiomyopathies. The symptoms vary by patient demographics and underlying causes, with risk factors including high blood pressure, high cholesterol, smoking, obesity, and genetic predisposition. Misdiagnosis is a complex challenge in CVD care, with significant rates of missed or incorrect diagnoses documented even in specialist settings. Technologies such as coronary calcium scans or CT angiography are not always available or suitable for all patients. Given the scale of the CVD burden and the variability in individual presentations, better predictive tools are needed.

Machine learning has already transformed diagnostic practices in numerous specialties, from radiology to oncology. ML models, particularly designed for classification, can analyse large amounts of patient data, detecting complex, non-linear patterns that may be missed in traditional statistical approaches or even human experts. Deep learning allows for the interpretation of medical images, while more classical algorithms are effective for structured, clinical datasets. Multiple studies have examined ML for heart disease prediction. For example, KNN has been highly effective, while others found ensemble or hybrid models. Hybrid models combining neural networks and classical models, outperformed single model systems. (Shah et. al, 2019) Across the literature, two factors stand out: the selection and optimisation of features, and the careful tuning and validation of models for reliable results.

Despite these successes, real-world deployment has challenges. Including inconsistent data quality, privacy concerns, and the black box nature of many model. Which can reduce clinical trust in automated predictions. Ensuring compliance, integrating with existing workflows, and continually validating models in new contexts is important for ML based diagnostic tools.

Dataset Description

The dataset covers both demographic and clinical variables, with the target variable being a binary indicator representing the presence or absence of heart disease.

Variable	Descripton
Age	Age of patient in years (Years)
Sex	Sex of the patient
Chest Pain	Chest pain type
Resting Blood Pressure	Resting blood pressure on admission to the hospital (mm Hg)
Cholesterol	Serum cholesterol (mg/dl)
Fasting Blood Sugar	FBS > 120 mg/dl (binary)
Resting ECG results	Resting electrocardiographic results
Maximum heart rate	Maximum heart rate achieved (bpm)
Exercise induced angina	Chest pain with exercise, Yes or No (Binary)
Old Peak	ST depression induced by exercise relative to rest
Slope	Slope of the peak exercise ST segment

Exploratory data analysis reveals important data patterns. For instance, most patients are male and between 40 and 70 years old, with a peak around age 55. Visualisations such as pair plots and correlation heatmaps show that variables like age and maximum heart rate are correlated with heart disease, while features such as cholesterol and resting blood pressure present more ambiguous relationships. Key risk indicators identified include high fasting blood sugar, flat or downward ST slope, and specific chest pain type.

Methodology

The methodology of machine learning pipeline follows for main phases, preprocessing of the data, feature engineering, train-test split and ensembles. Having multiple phases improves accuracy of the data as multiple different algorithms can be applied to compare performance. Beginning with preprocessing of data all raw data was cleaned, with special attention to encoding categorical variables and scaling continuous features. Transforming variables like chest pain type into several binary features. This allows for feature engineering, where, variables most likely to influence outcomes are prioritised, with lesser features identified and potentially dropped through model evaluation rounds. A train-test split separates 75% of the data to use in training the models, while the remaining 25% is left for model testing and

evaluation. With the training set, five models were chosen to balance interpretability and predictive strength. Support Vector Machines (SVMs) isolates two classes of data which is chosen due to its ability to separate the group with CVD to those who do not. K-Nearest Neighbours (KNN) was used due to its simplicity in handling multiclass and non-linear problems. The model's only hyperparameter is k , the number of neighbours. Naive Bayes was selected for its computational efficiency and effectiveness, especially when features are independent. It serves as a strong baseline in classification tasks with mixed feature types. Decision trees use regression methods to classify the data based on the value of the variable. This classifies data accurately with as minimal depth as possible. Logistic regression is used to model one or more predictor variables, with the ability to detect non-linear relationships within the data it can calculate the odds of a patient having CVD based on their attributes.

Based on these methods, the models can be further trained with hyperparameter tuning and cross-validation. Parameter settings such as the value of k in KNN or the tree depth for decision trees can be optimised to achieve maximum accuracy based on the results of the initial method on the test set. Models can also be combined to improve accuracy further. By using an ensemble voting classifier, several base models (SVMs, KNN, Naive Bayes, Decision trees and Logistic Regression) predictions are aggregated from the top-performing individual models for greater reliability. This allowed for the quality of the model to be evaluated based on attributes and visualised, with important aspects being: The shape of correct predictions overall, class-specific evaluations for positive and negative outcomes, F1-score, measuring precision and recall within class distributions. Performance across multiple threshold settings such as Area Under the Curve (AUC), where 0.9 or higher indicates a highly effective model and detailed error analysis such as confusion matrix, mean squared error to determine probabilistic calibration.

Results

In evaluating ML algorithms for CVD predictions, the models show competitive performance. Logistic regression and naive bayes are the top performing classifiers, both achieving an accuracy rate of 0.865 and strong cross validation results, suggesting their predictive ability generalises well to unseen data. These models also recorded high AUC values of 0.94, indicating capability in distinguishing between patients with and without CVD. KNN and SVMs were also reliable, with scores of 0.865 and an AUC of 0.93. Decision Trees performed lower with an accuracy of 0.804, primarily due to overfitting on the training data. Cross-validation played a central role in the thesis, providing a more robust assessment than simple train-test splits. The minor differences between accuracy and cross-validation scores for Logistic Regression, Naive Bayes, KNN and SVM highlight that these models are less prone to overfitting and can be trusted for new patient predictions. In contrast, Decision Tree's wider gap between training and cross validation accuracy flags potential unreliability in real world use.

The Voting Classifier leveraged the strengths of individual models for improved results. By combining predictions, this method boosted overall accuracy to 0.878, outperforming any single model and providing added robustness and reliability. The voting classifier's results demonstrate how ensemble learning can mitigate individual model weaknesses and ensure more consistent clinical support. Feature importance investigations pinpointed the variables most important in the models. Consistently, features such as cholesterol, chest pain type, fasting blood sugar, sex, slope, exercise induced angina, and occasionally maximum heart rate ranked highest in driving the model's predictions. Notably, commonly expected features like age or resting ECG results had limited impact, emphasising the need for objective, data-driven feature selection.

Practical Implications in deployment

Robust real-world deployment requires more than just training a high accuracy model. The thesis explains the prediction workflow, from encoding and scaling new patient data, through application of the voting classifier, to generation of reliable predictions. Encoding and preprocessing steps were critical for effective deployment. These steps guarantee that input features match the training format, minimising errors arising when models are integrated for clinical data sources.

The thesis highlights how a dependable, automated workflow covering everything from feature transformation to prediction export forms the backbone of clinical integration. Only with strict reproducibility and attention to operational details can ML models be trusted by healthcare professionals for real time decision support.

Limitations and Recommendations

While the combined dataset is larger than many used in similar studies, its sub 1000 sample size restricts generalisability. Larger, more diverse datasets are required for production level models, as limited data can artificially inflate accuracy and reduce stability across patient populations. The performance of each classifier depended heavily on selected features and the tuning of hyperparameters. Variations in encoding or scaling can significantly affect these results, emphasising the necessity for rigorous model retraining when applied in new environments or with different data sources.

The thesis demonstrates the power of basic demographic and clinical data, but also acknowledges the potential for dramatic improvement by adding more variables such as genetic, imaging, longitudinal tracking, and real time biosensor data. The recommendation is to expand feature sets and periodically revalidate models for evolving populations.

Conclusion

Machine learning stands as a powerful tool in the early detection and risk stratification of cardiovascular disease. The thesis convincingly demonstrates the value of ML methods, especially ensemble approaches, in achieving strong and generalisable predictive

performance on real clinical data. With clear documentation of challenges, workflow, and integration, key insights are offered in both the technical and practical sides of model deployment.

References

Hossain, J. (2024). A comparative analysis of machine learning techniques and key insights for cardiovascular disease prediction. *Theseus.fi*.

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